

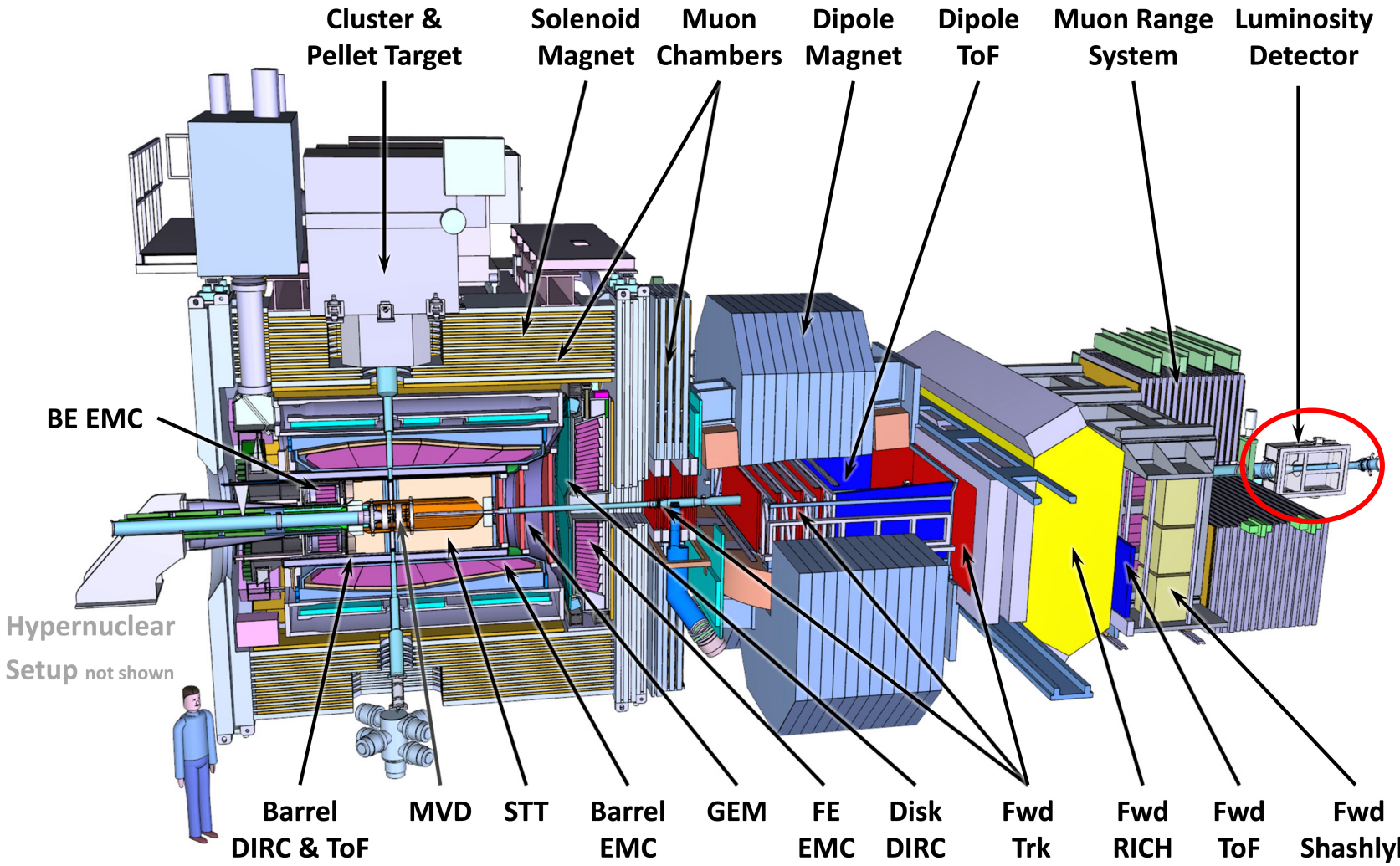
Luminosity Determination with Restgas Target

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PANDA Detector



Why to Measure Luminosity

$$N = L \cdot \sigma$$

Number of Events

Time-integrated Luminosity

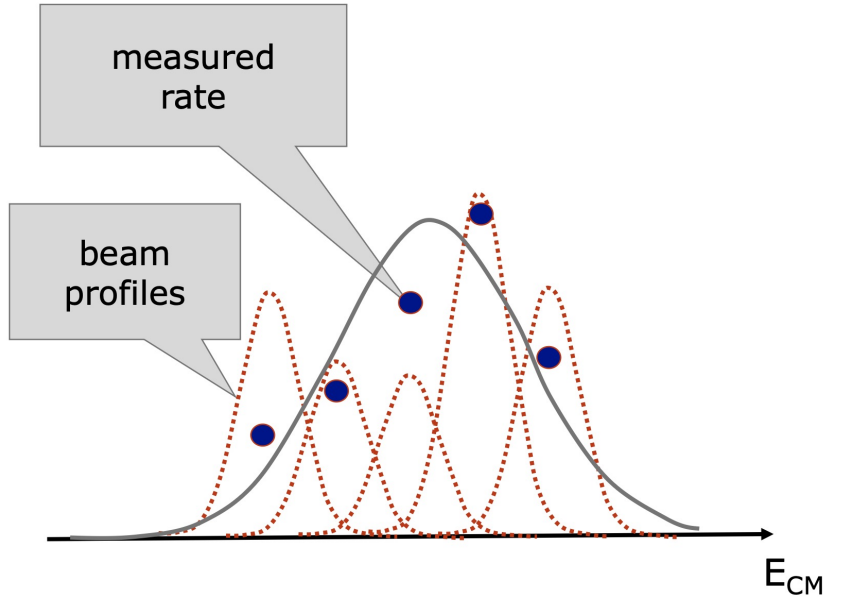
Cross Section

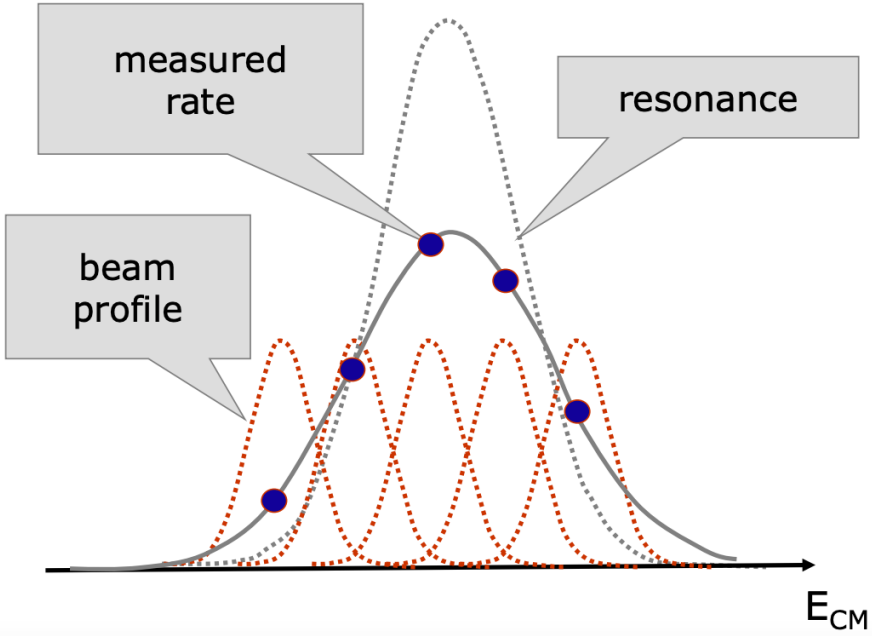
- Absolute cross sections

⇨ Absolute time-integrated luminosity

- Scan experiments

⇨ Relative time-integrated luminosity





How to Measure Luminosity

⇨ $\bar{p}p$ elastic scattering used, only $\bar{p}$ measured

$$\frac{dN(\theta)}{d\theta} = L \cdot \left[\frac{d\sigma(\theta)}{d\theta} \cdot \epsilon(\theta) \right] \otimes Res(\theta)$$

Number of Events

Time-integrated Luminosity

Differential Cross Section

Reconstruction Efficiency

Detector Resolution

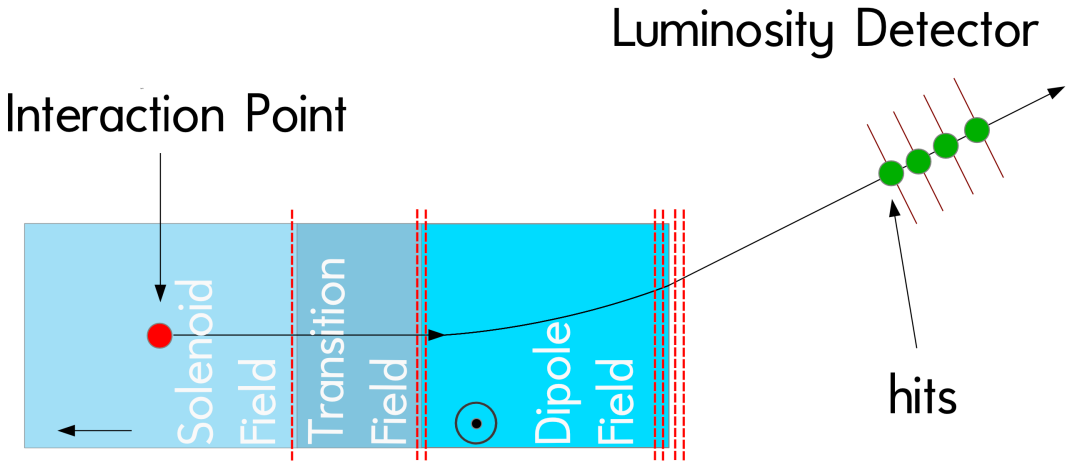
⇨ Determine efficiency and resolution

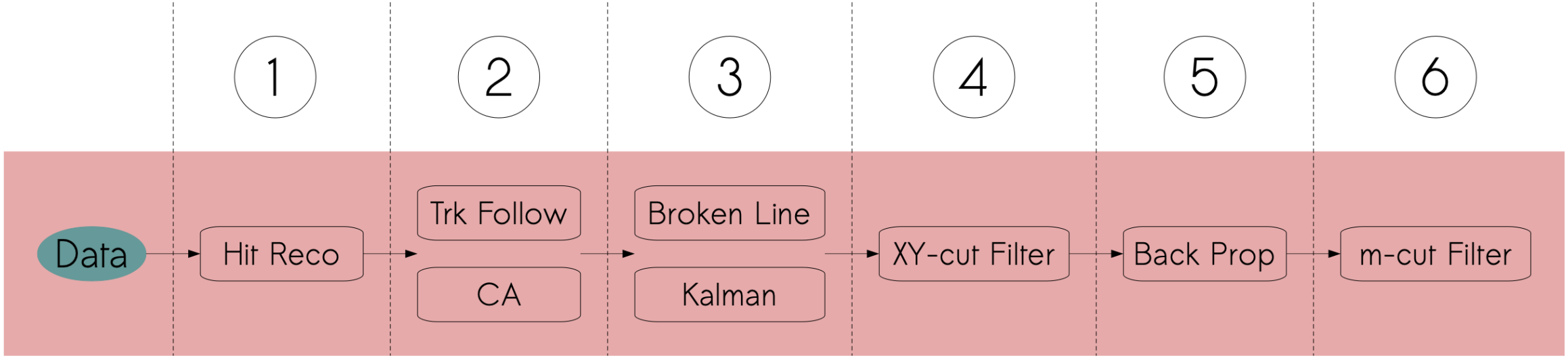
⇨ Determine Luminosity by comparing data and model

Track Reconstruction Software

(PandaRoot/macro/detectors/lmd)

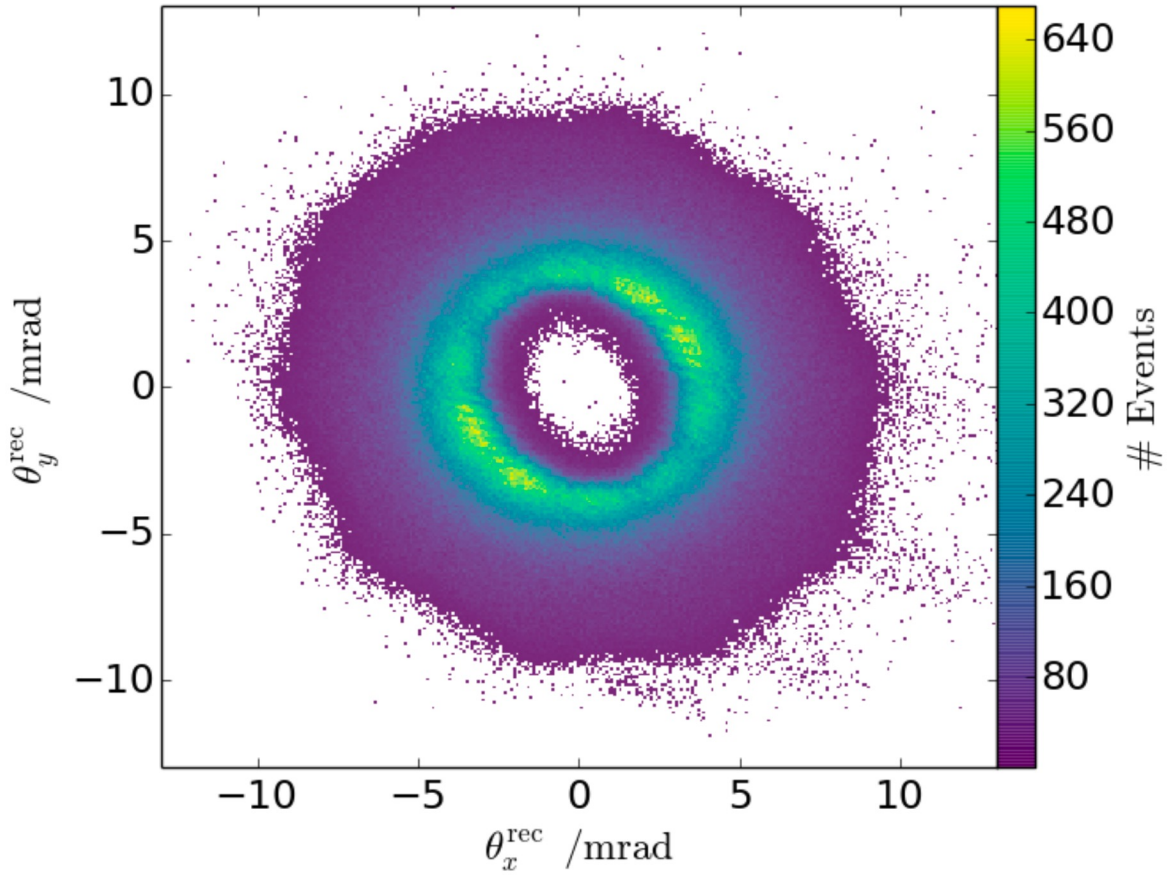
- Sensor data





Track Reconstruction Software

- Final Data → Distribution of tracks at IP

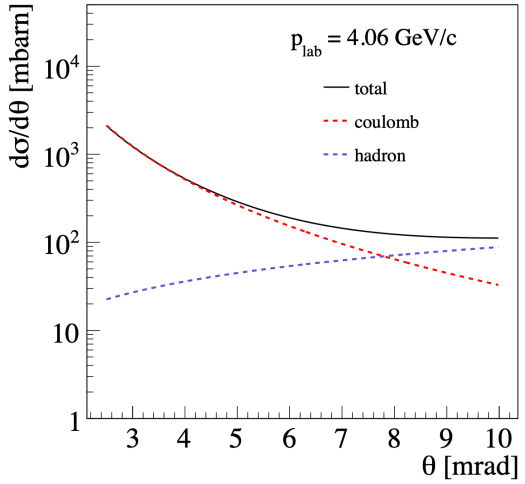


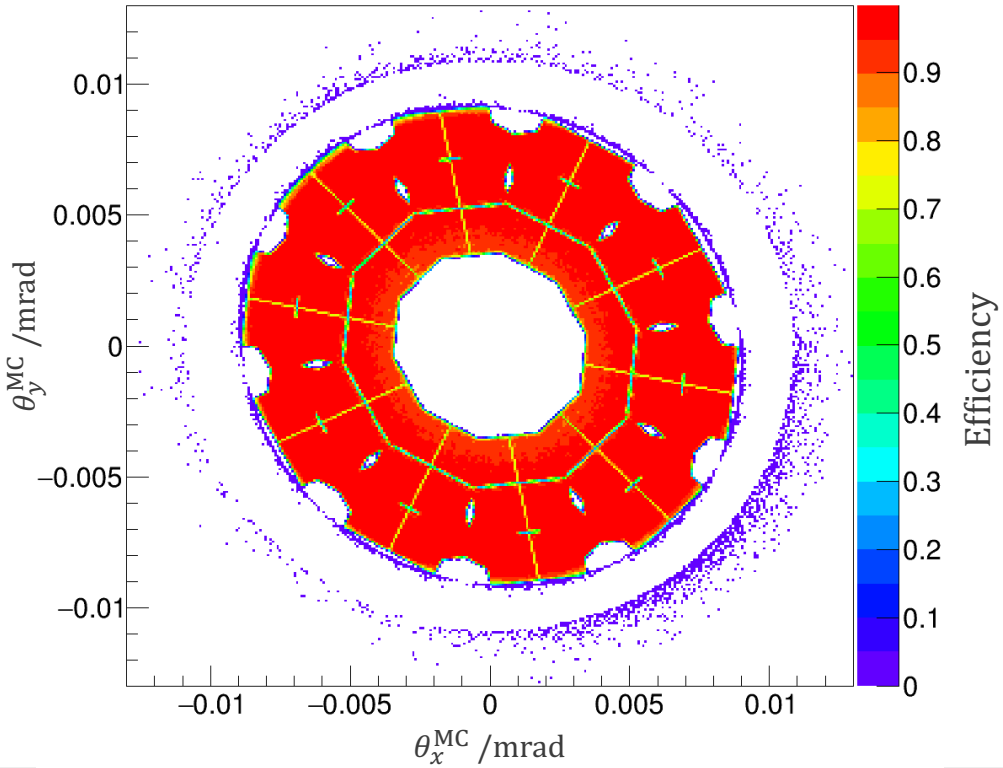
Luminosity Fit Software

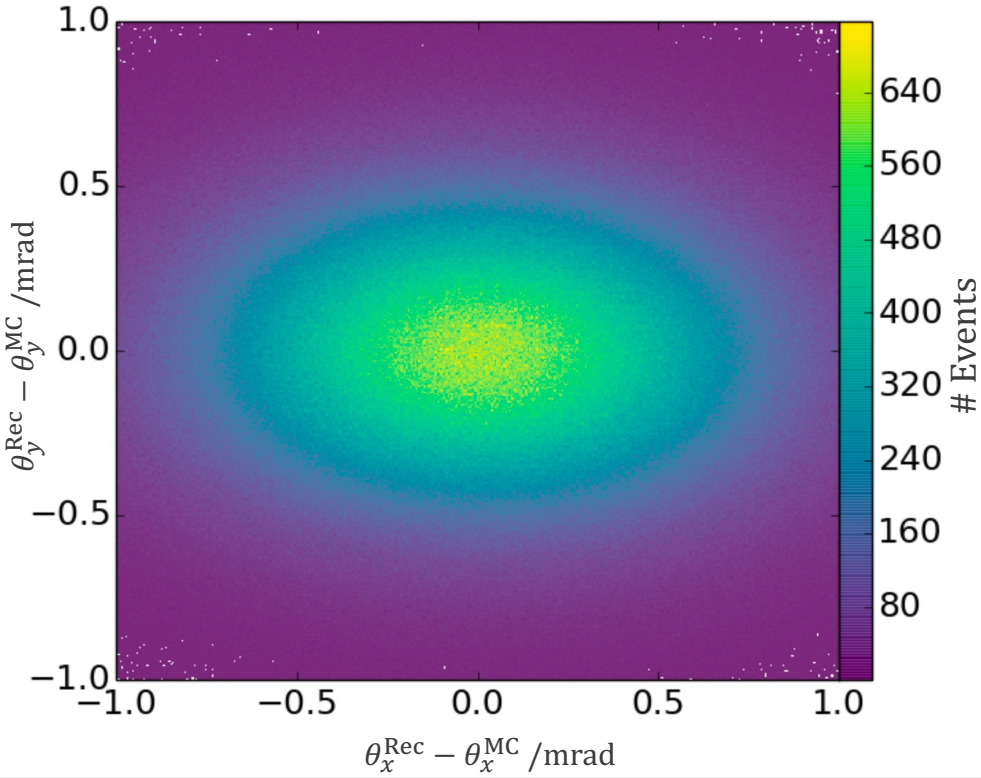
(github.com/LuminosityFit)

- Preparation of the fit model

⇨ Lmd Efficiency & Resolution corrections

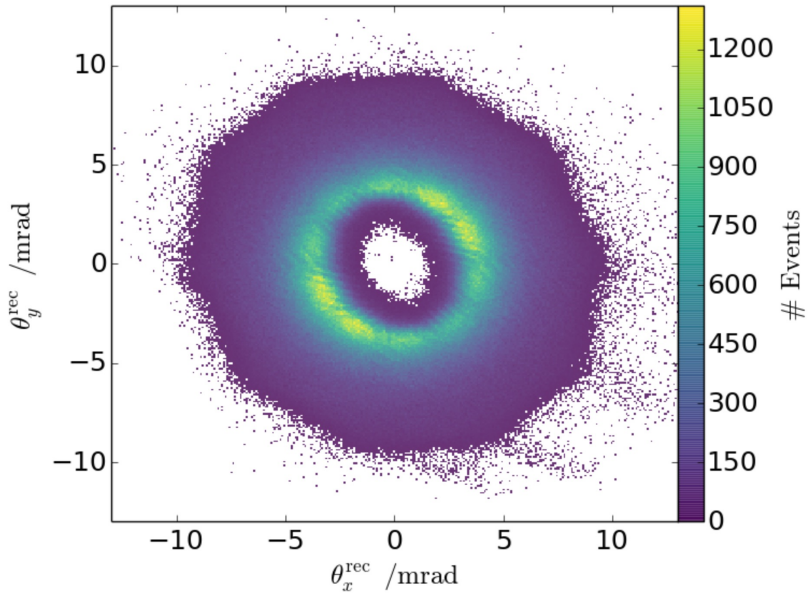






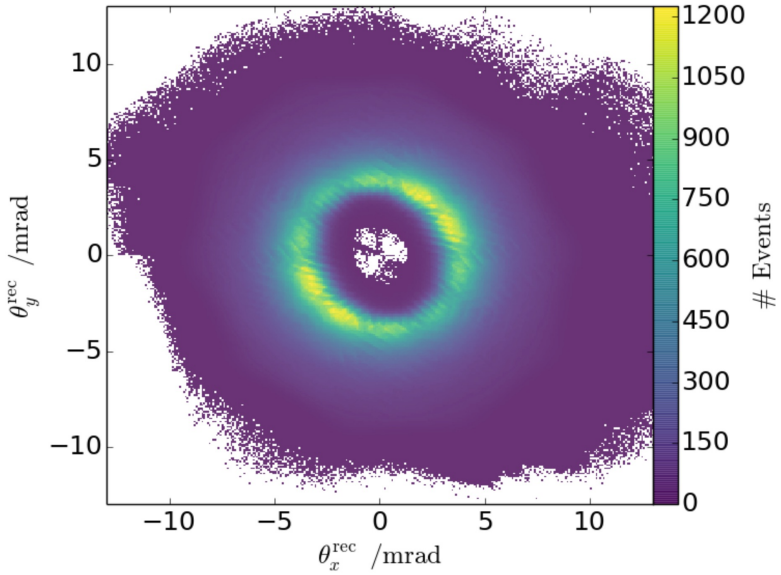
Luminosity Fit Software

- Fit MC model to data distribution



Reconstructed data:

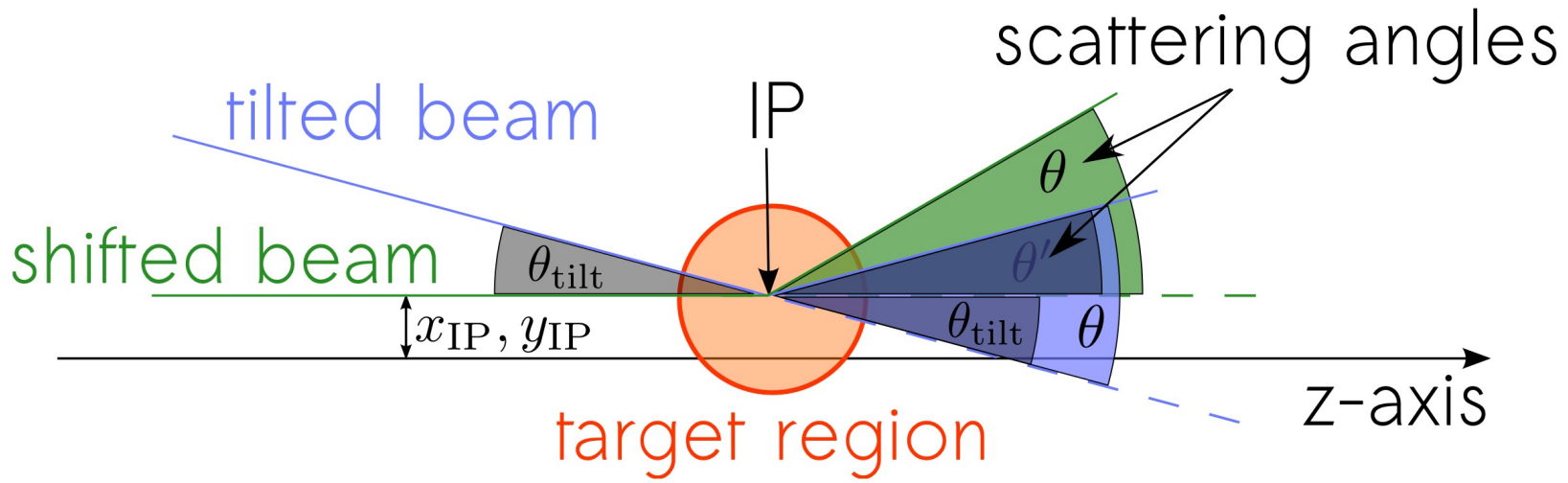
- Back propagated to IP
- Filtering



Fit model:

- Based on elastic cross section
- Corrected with efficiency and resolution

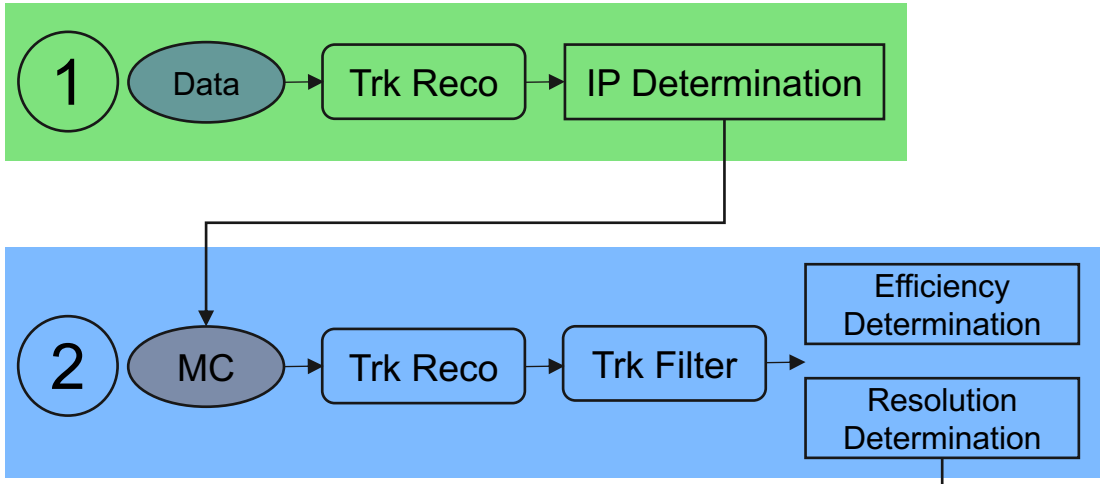
Luminosity Fit Software —— Complications

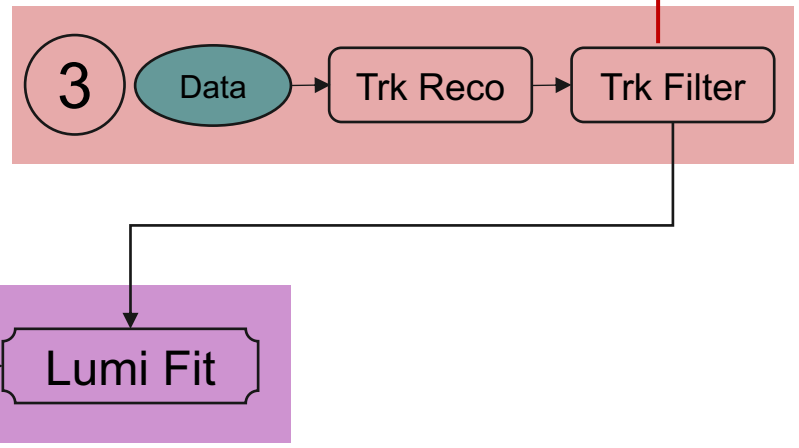


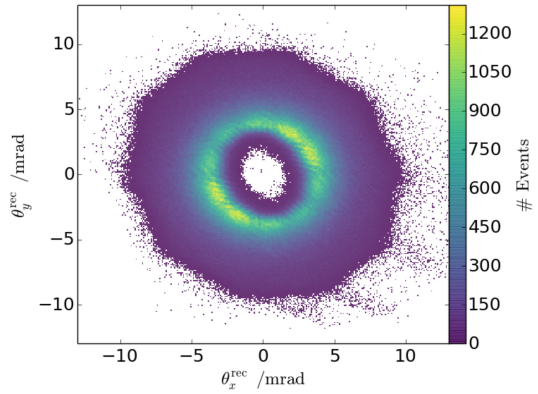
- determine IP ⇒ efficiency & resolution correction
  - IP position: measured by reconstructed tracks (only x, y)
  - IP distribution: taken from target and accelerator group
- determine beam corrections
  - tilt
  - divergence

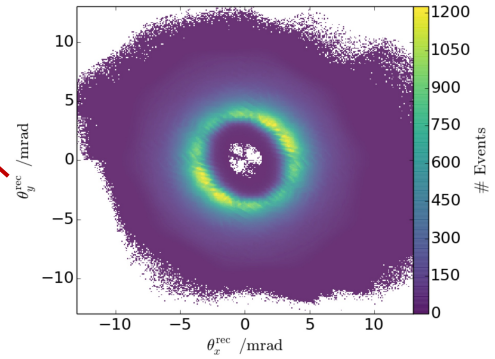
} Determined by LuminosityFit Software

Luminosity Fit Process







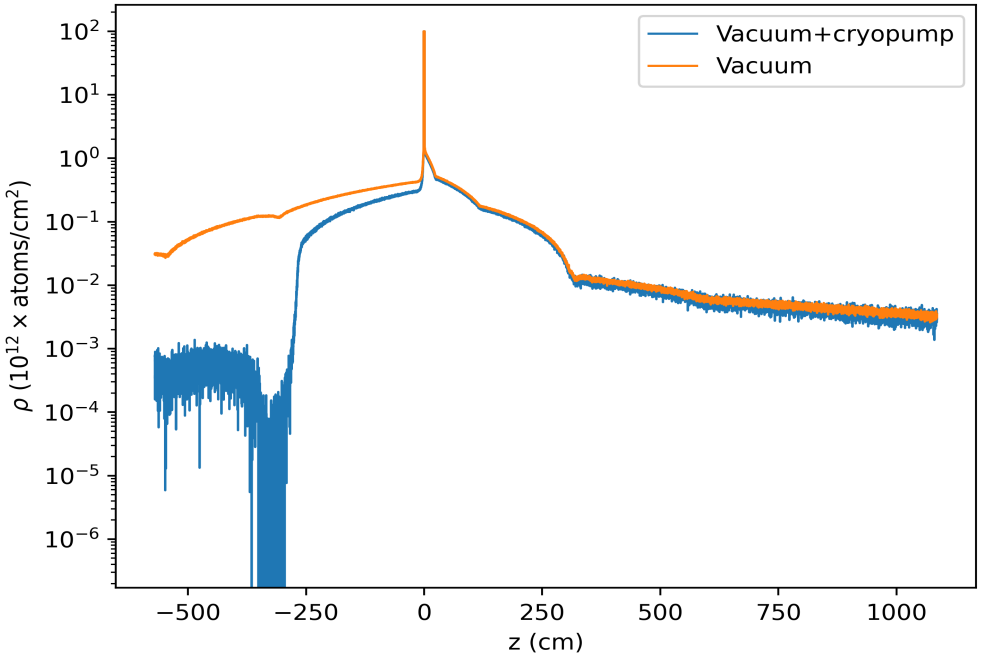


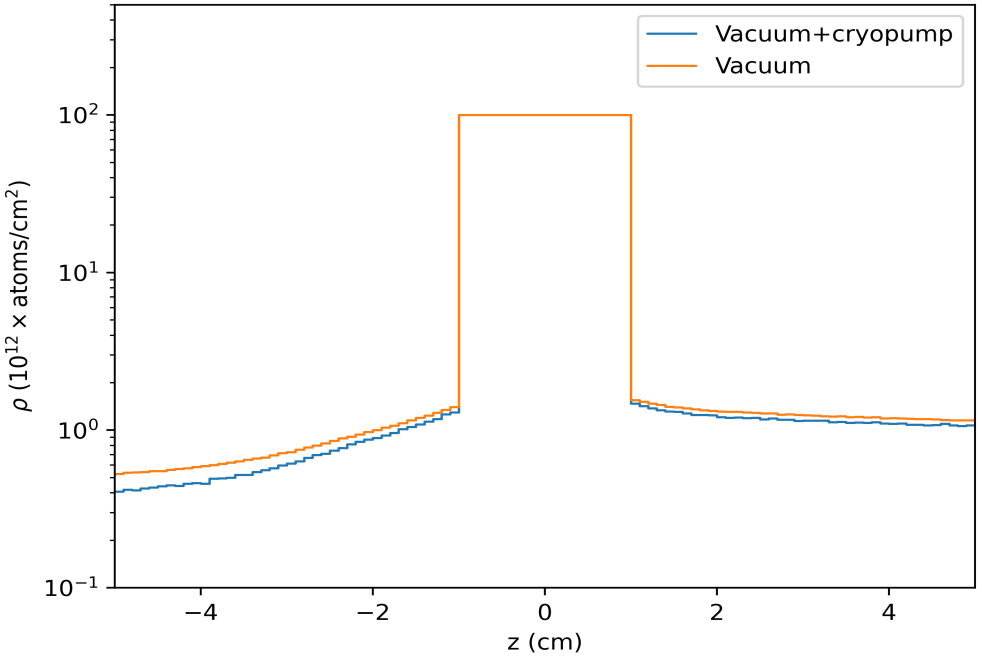
Output:

- Time-integrated Luminosity
- Tilt and divergence of the beam

Restgas Distribution

From target group in Muenster





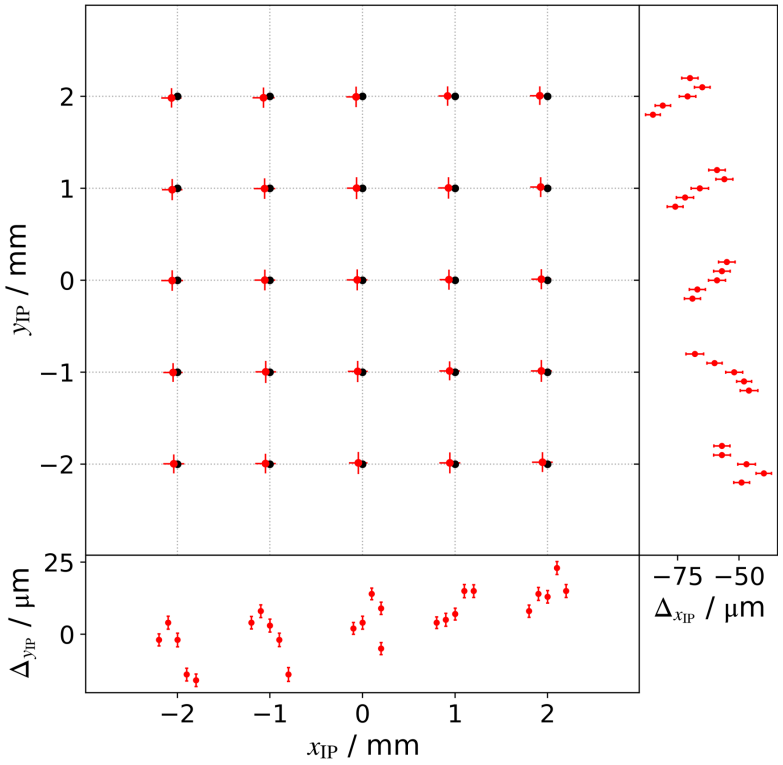
Luminosity detector

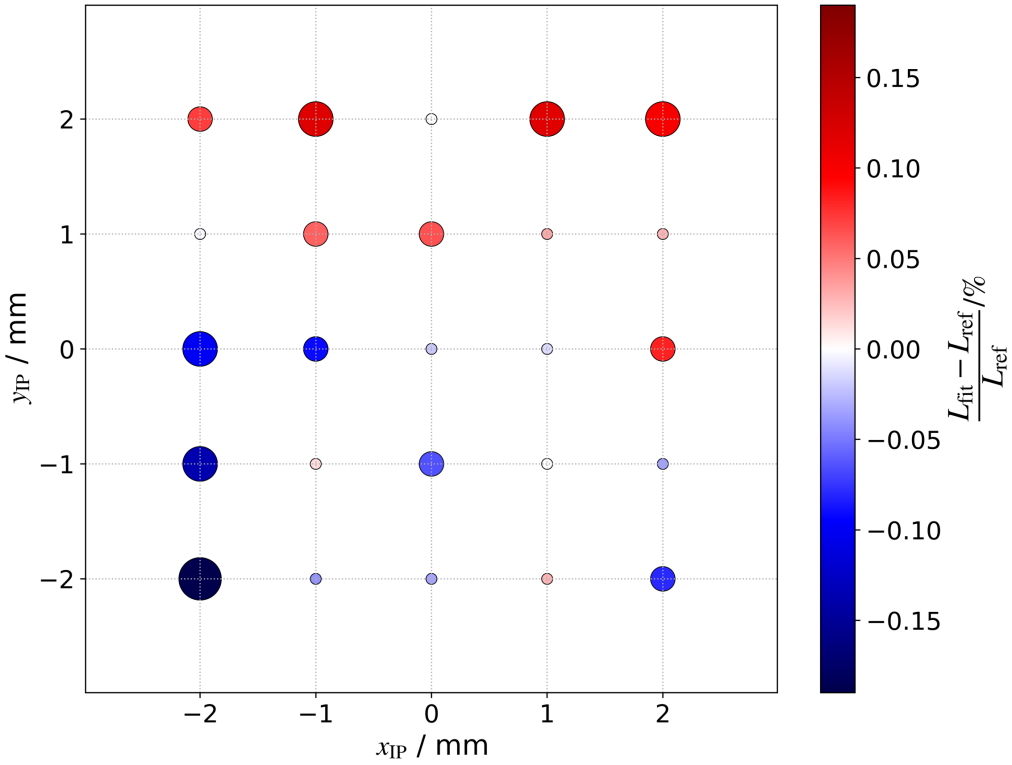
- Track reconstruction not sensitive to z position of the origin of the tracks

Check with the Restgas Target

Scenarios with IP shift / beam tilt / beam divergence only

- Determination of IP shift $(p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c)$





⇨ IP shift:
$|\Delta x| \leq 85\ \mu\mathrm{m}$
$|\Delta y| \leq 25\ \mu\mathrm{m}$
$|\Delta L/L| < 0.20\ \%$

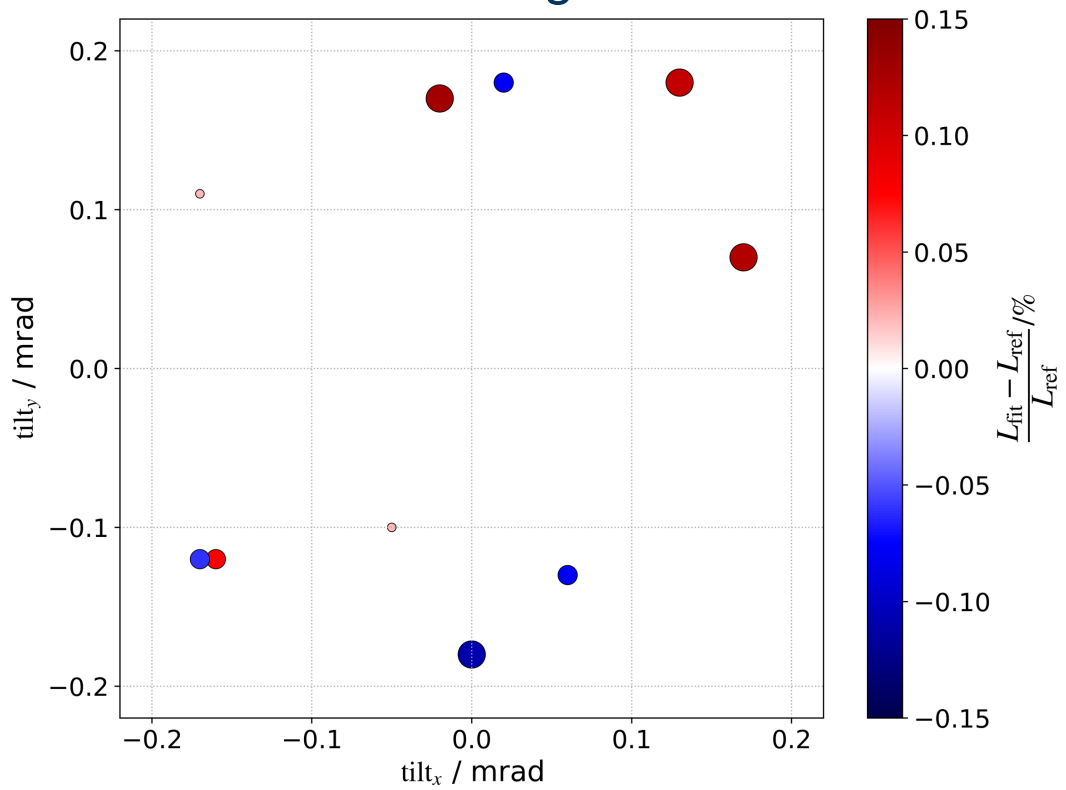
⇨ Beam tilt:
$|\Delta\theta_x| \leq 7\ \mu\mathrm{rad}$
$|\Delta\theta_y| \leq 9\ \mu\mathrm{rad}$
$|\Delta L/L| < 0.11\ \%$

⇨ Beam divergence:
$|\Delta \mathrm{div}_x| \leq 40\ \mu\mathrm{rad}$
$|\Delta \mathrm{div}_y| \leq 45\ \mu\mathrm{rad}$
$|\Delta L/L| < 0.70\ \%$

Comparison w/o Restgas (Realistic Scenarios $p = 4.06\ \mathrm{GeV}/c$)

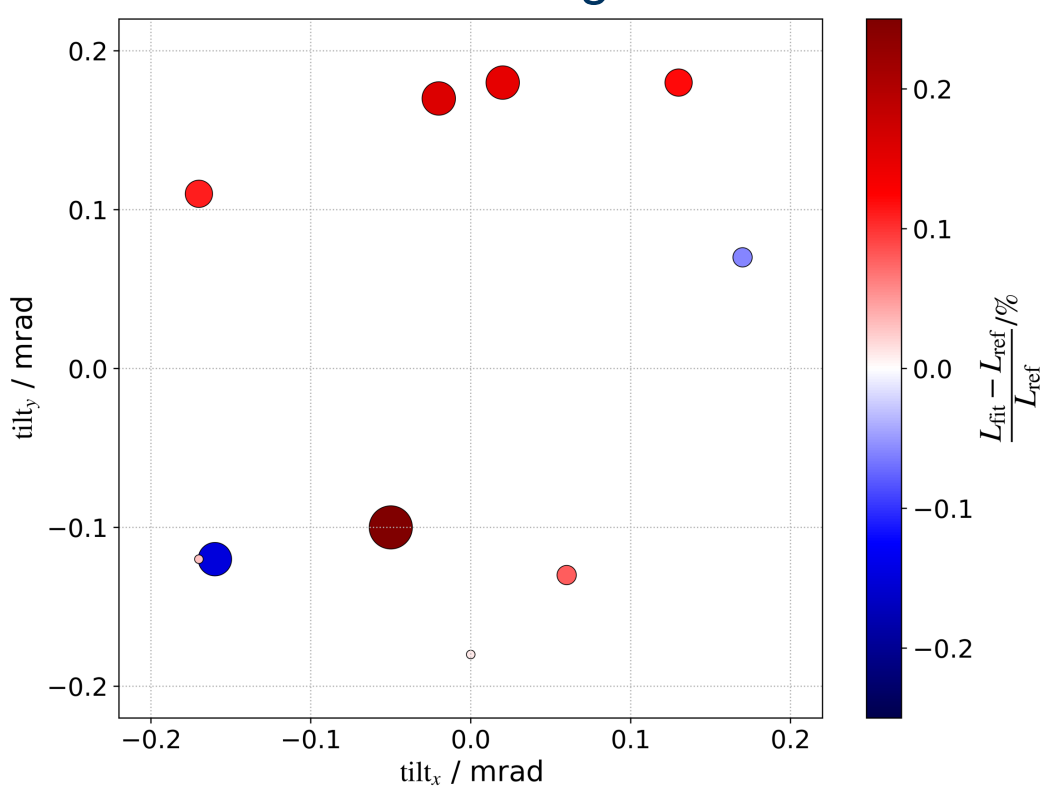
Luminosity

With Restgas



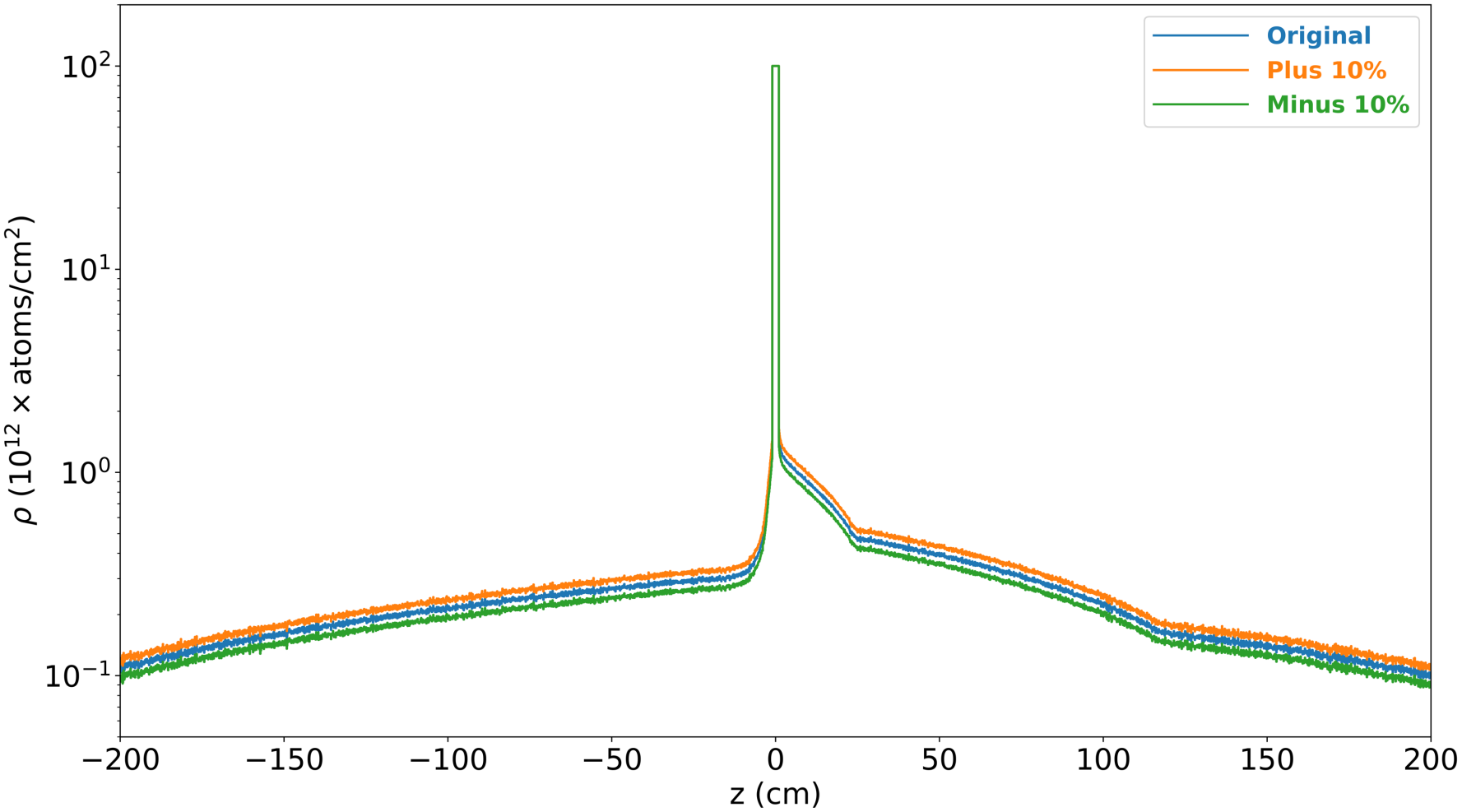
⇨ Maximum deviation:
$|\Delta L/L| \leq 0.15 \pm 0.17\%$

Without Restgas



⇨ Maximum deviation:
$|\Delta L/L| \leq 0.25 \pm 0.08\%$

Luminosity Fit Results with Different Restgas profiles



Luminosity Fit Results with Different Restgas profiles

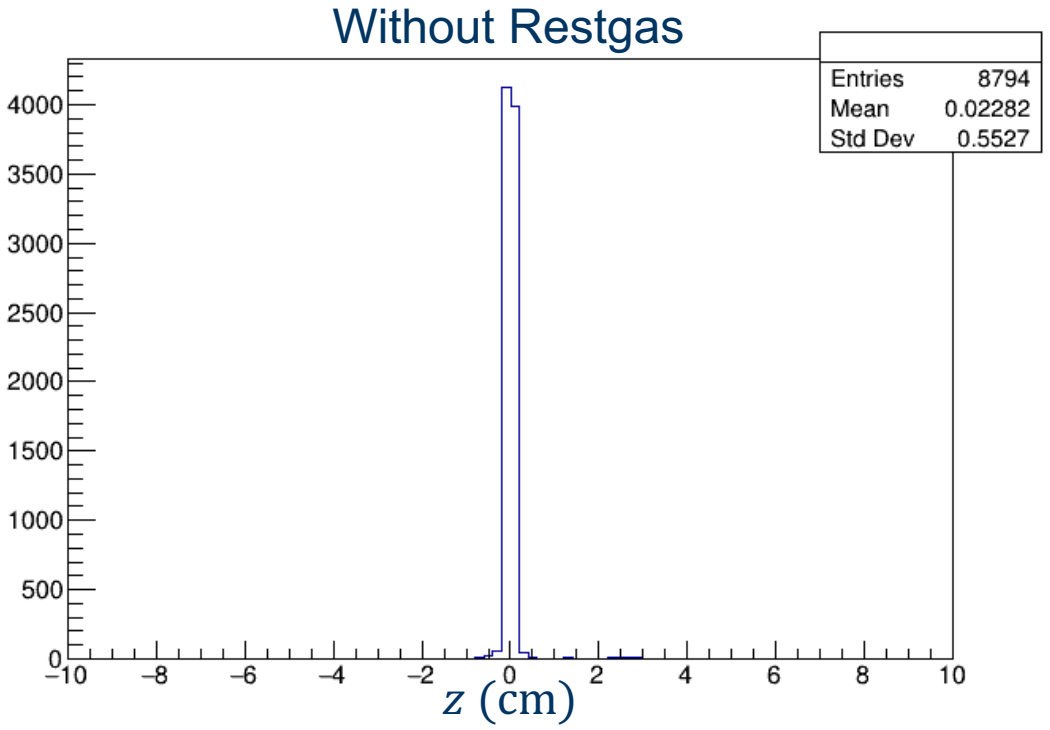
- $p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c$
- IP cut around reconstructed IP: $x, y < 1\ \mathrm{cm},\ z < 5\ \mathrm{cm}$
- Corrected generated luminosity by $\mathrm{ratio} = \dfrac{\text{Number of MC events after IP cut}}{\text{Number of generated MC events}}$

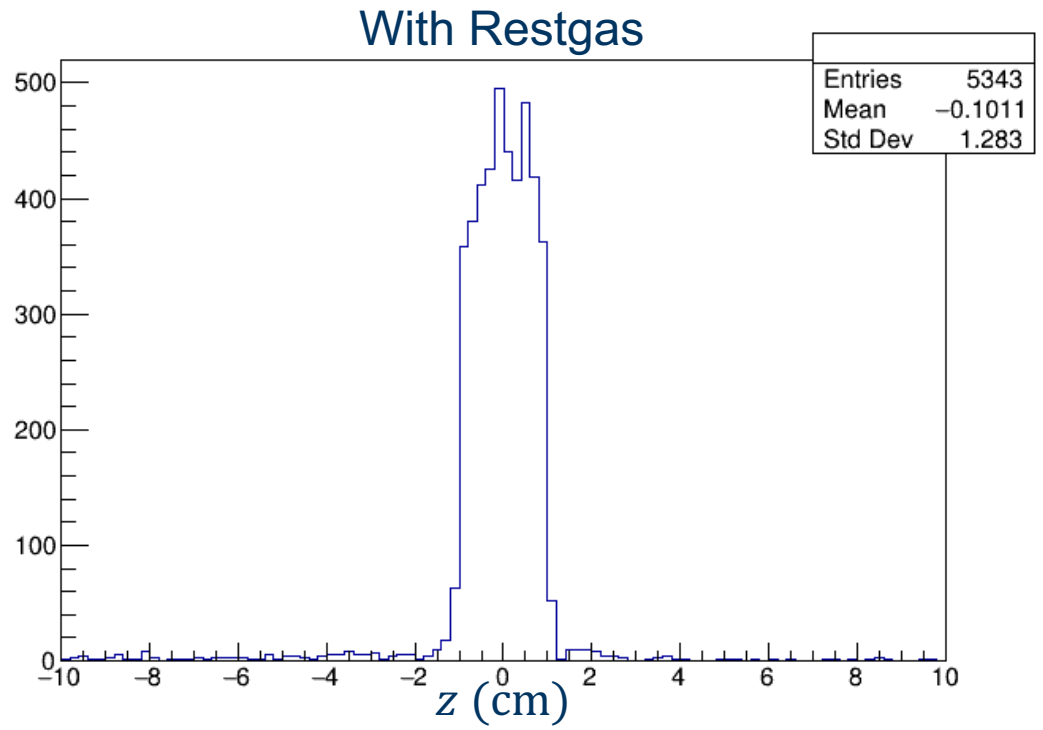
Restgas Profiles	Generated Luminosity	Corrected Generated Luminosity	Measured Luminosity	Corrected $\Delta L/L$ (%)	$\Delta L/L$ with Original Restgas (%)
Original	3.495×10^6	2.263×10^6	2.262×10^6	0.016 ± 0.08	0.016 ± 0.08
Plus 10%	3.513×10^6	2.354×10^6	2.354×10^6	-0.018 ± 0.08	-3.39 ± 0.08
Minus 10%	3.495×10^6	2.187×10^6	2.188×10^6	0.016 ± 0.08	3.39 ± 0.08

⇨ Luminosity quite good with different restgas profiles

Restgas Distribution Determination

- ⇨ $\bar{p}p$ elastic scattering with restgas
- ⇨ Reconstruct $\bar{p}$ and p with the whole PANDA
- ⇨ Determine IP after vertex fit





Restgas Distribution Determination

⇨ $\bar{p}p$ elastic scattering with restgas

⇨ Reconstruct $\bar{p}$ and p with the whole PANDA

⇨ Determine IP after vertex fit

To do next:

⇨ Develop tracking algorithms for tracks not from $(0, 0, 0)$

⇨ Check reconstruction efficiency dependence on IP in z

⇨ Apply efficiency correction on restgas distribution

⇨ Determine precise restgas distribution

Summary and Outlook

- Study with Restgas at $p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c$:

	Without Rest Gas	With Rest Gas
$\lvert\Delta L/L\rvert$	$< 0.25 \pm 0.08\ \%$	$< 0.15 \pm 0.17\ \%$

- Cutting around IP is applied in Luminosity Fit

⇨ Relative deviation of Luminosity: $< 0.5\%$ for p_{lab} $1.5\mathrm{GeV}/c - 15\mathrm{GeV}/c$

⇨ Next Steps:

- Develop tracking algorithms
- Determination of reconstruction efficiency
- Precisely restgas distribution determination with $\bar{p}p$ elastic scattering



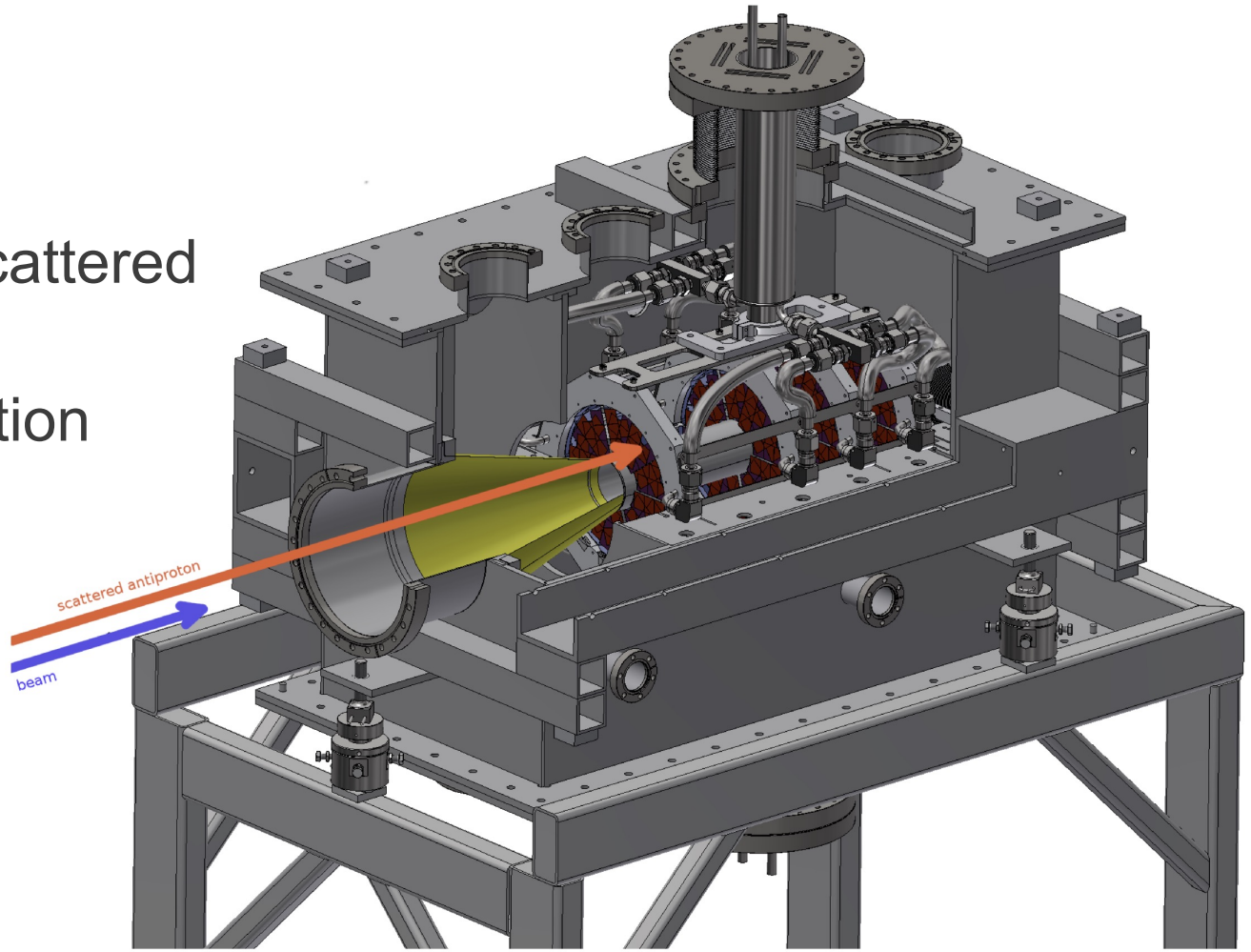
Thanks For Your Attention!

Backups

Luminosity Detector

Measurement of the forward scattered $\bar{p}$ with 4 planes of HV-MAPS

→ 4 points for track reconstruction

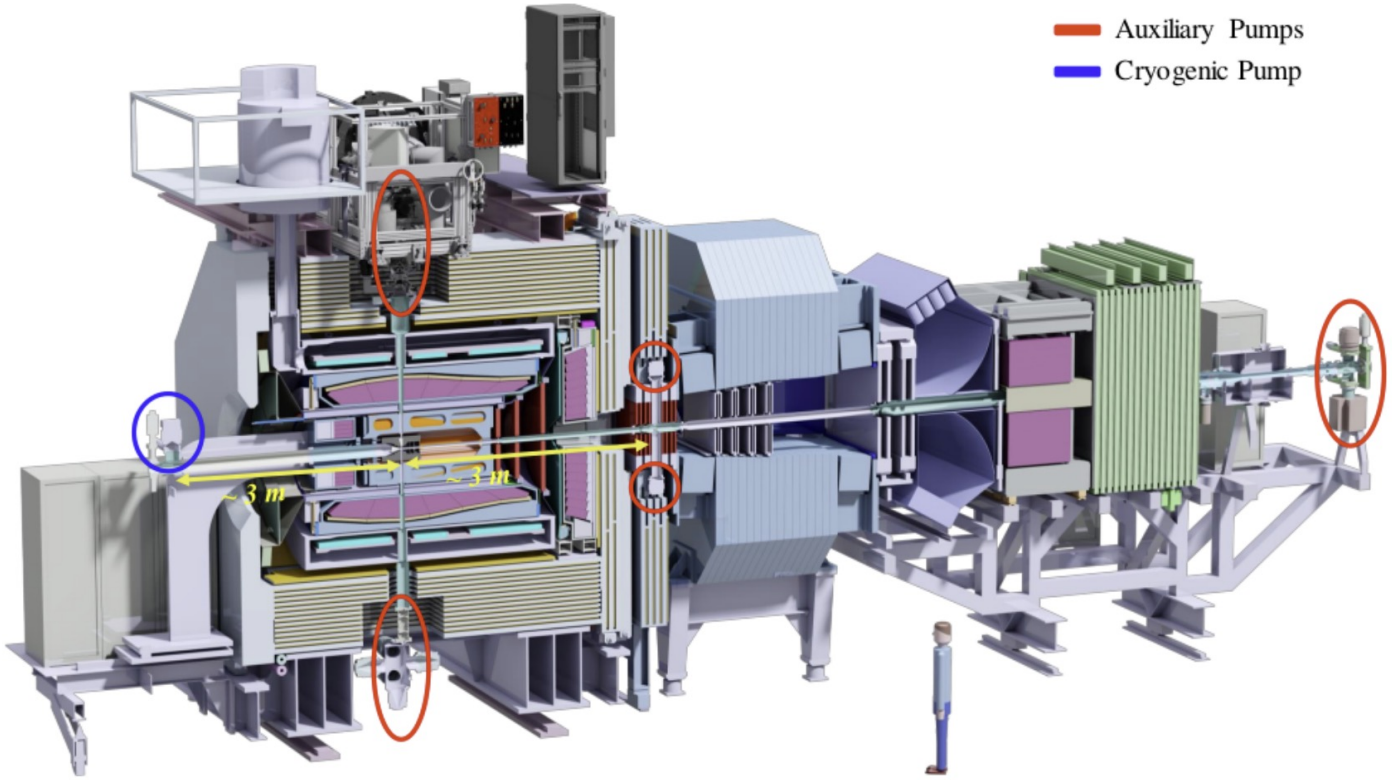


Restgas in Beam Pipe

Dissipation of target into beam pipe

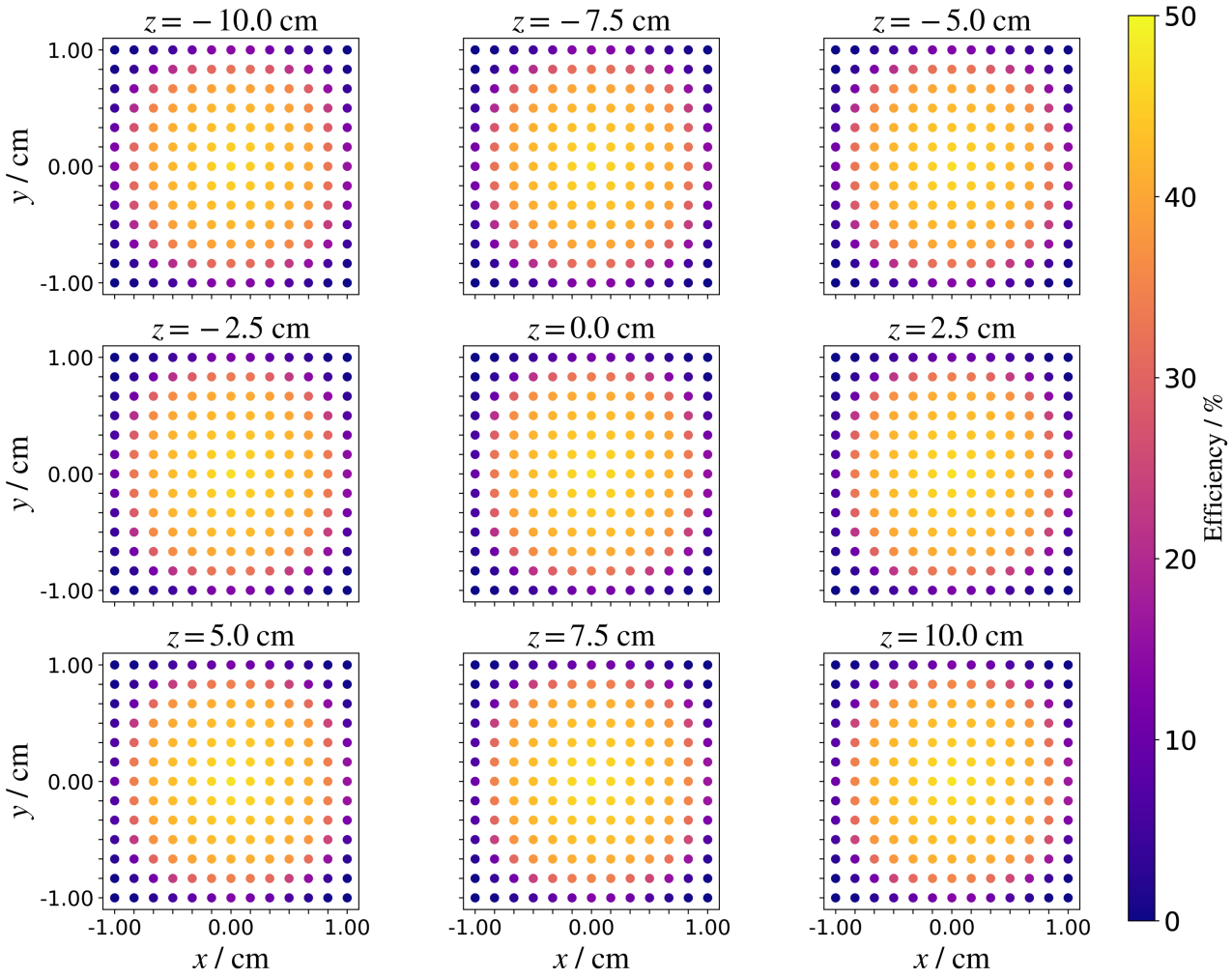
- H - cluster evaporation
- Vacuum pumps far away

⇒ Restgas distribution

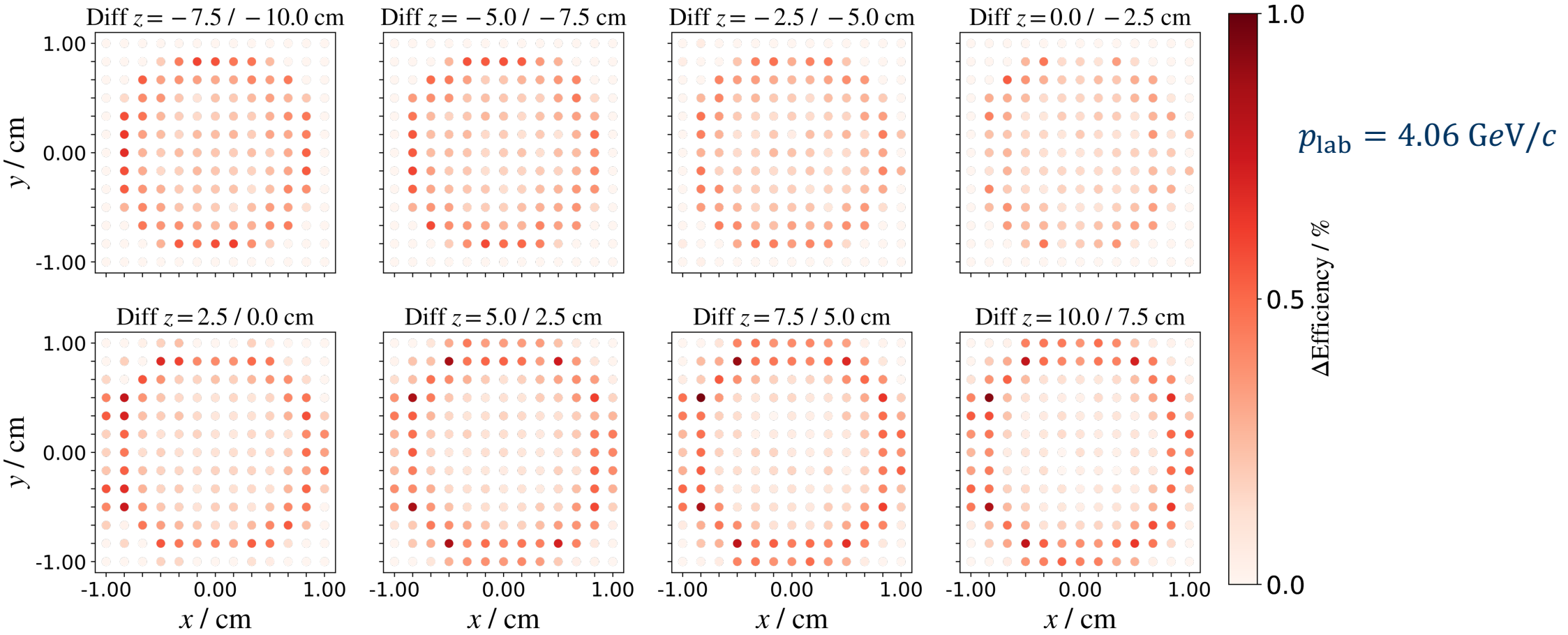


Efficiency Dependent on IP

$p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c$



Efficiency Difference Between Two z Positions



⇨ Efficiency increase with z

Check of Fit Procedure

steps:

- Include IP position
- Fit Parameters:
  - Beam tilt
  - Beam divergence
  - Luminosity

Analysis Procedure:

- Fix divergence ⟹ get tilt
- Fix tilt ⟹ get divergence
  ⟹ get luminosity

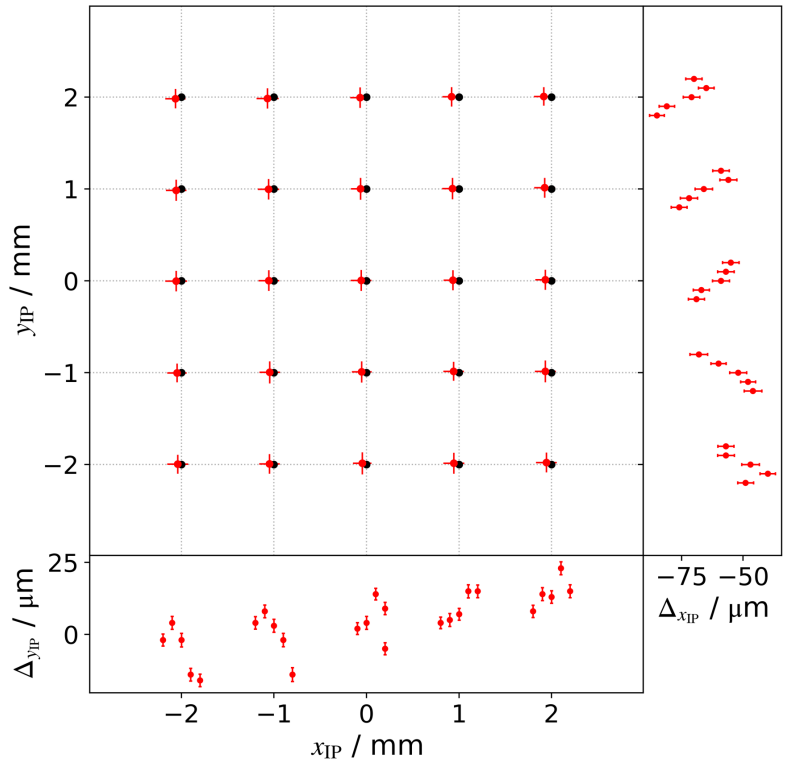
- Fit with divergence fixed
- $p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}$

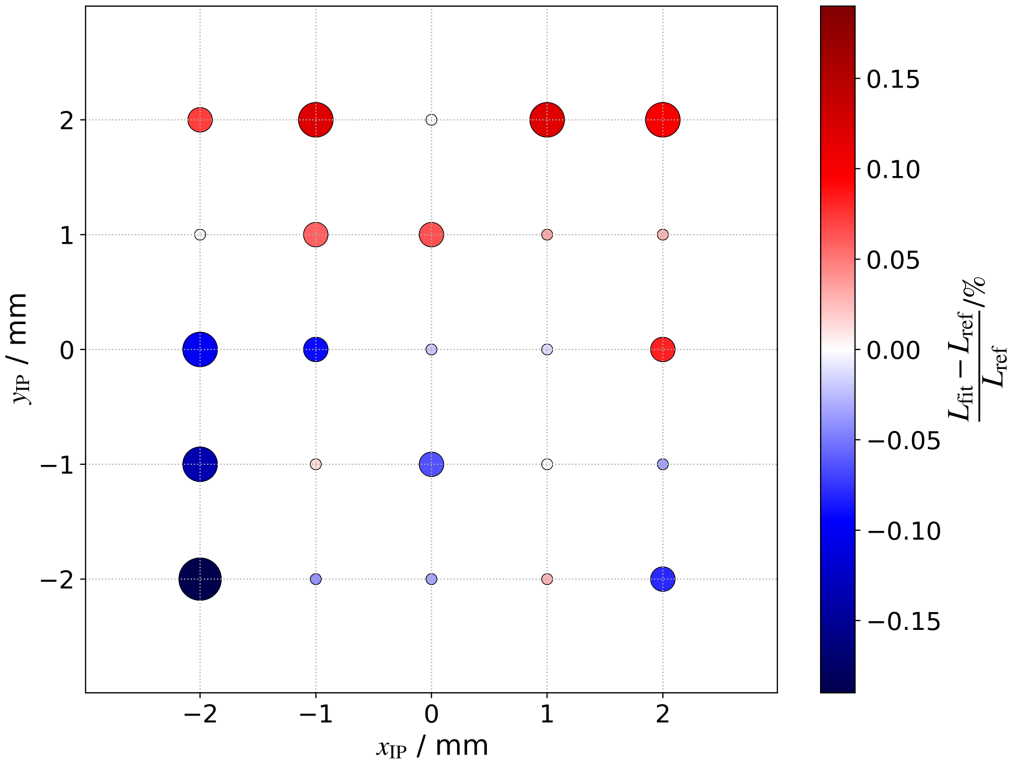
tilt_x	tilt_y (μrad)	div_x	div_y	$\Delta L/L$ (%)
100	100	100	100	0.30 ± 0.07
105	96.3	0	0	
100	100	100	100	-0.05 ± 0.07
105	95.9	100	100	
100	100	100	100	-1.05 ± 0.07
104	94.5	200	200	

Check with the Restgas Target

Scenarios with IP shift only

- Determination of IP shift $(p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c)$





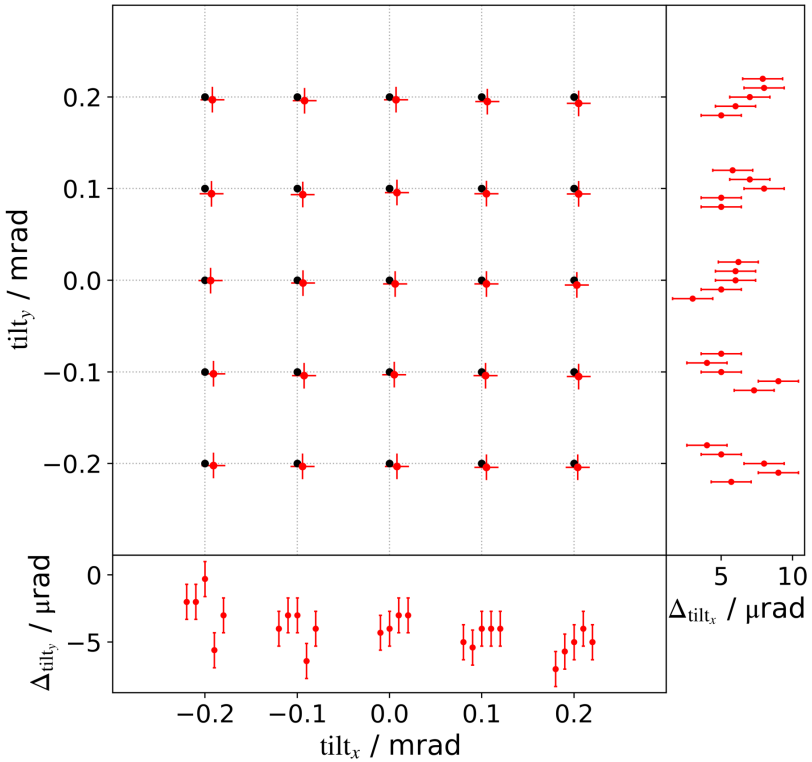
⇨ Maximum deviation after correction:

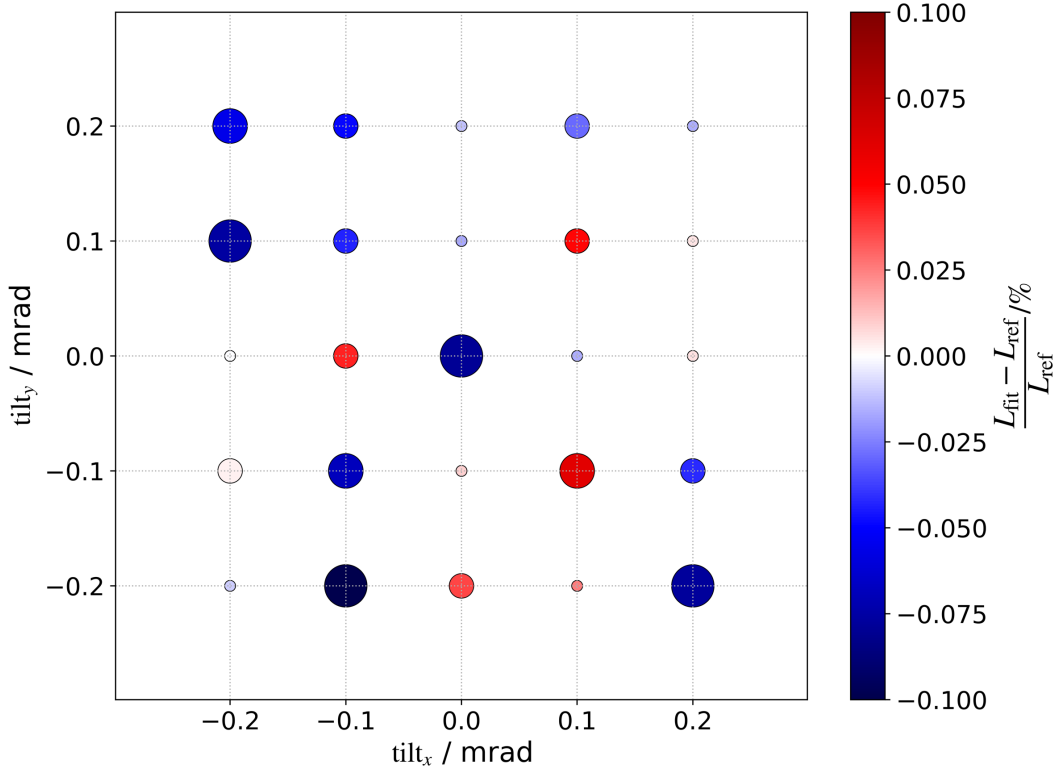
$|\Delta x| \leq 85\ \mu\mathrm{m}$ $|\Delta y| \leq 25\ \mu\mathrm{m}$ $|\Delta L| < 0.20\ \%$

Check with the Restgas Target

Scenarios with beam tilt only

- Determination of beam tilt $(p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c)$





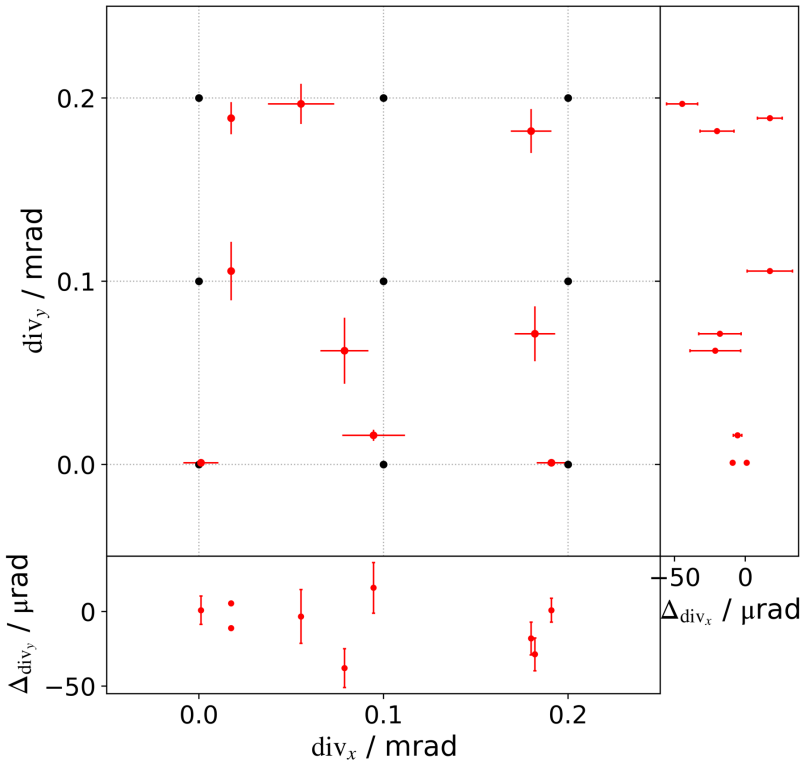
⇨ Maximum deviation after taking into account:

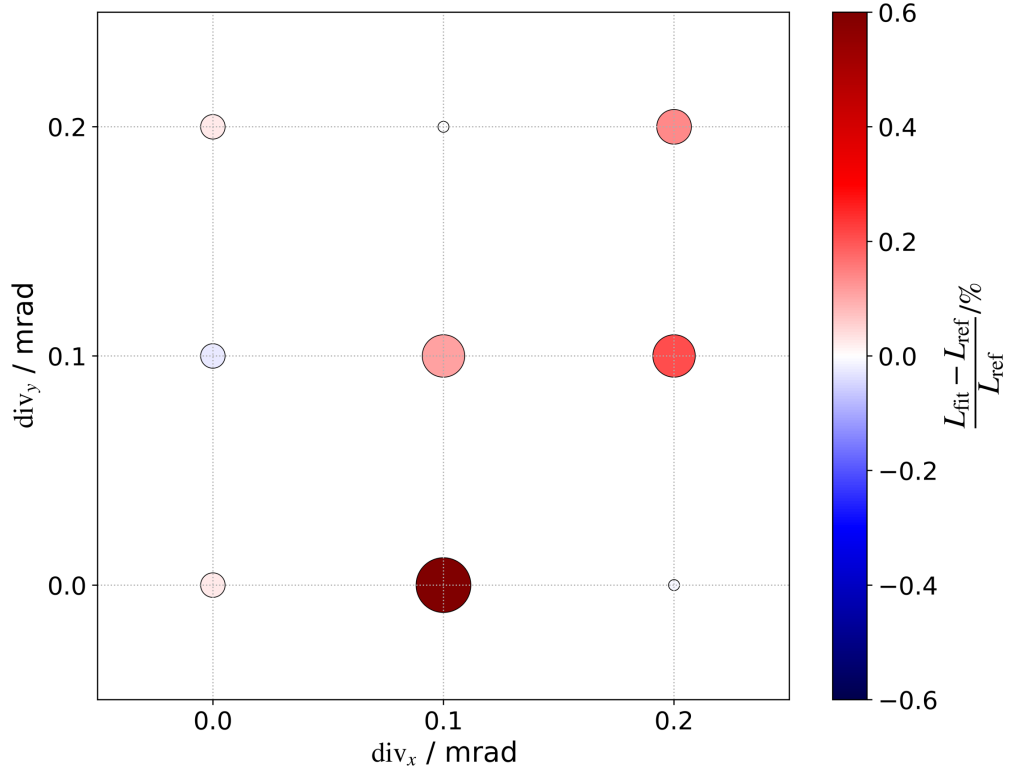
$|\Delta\theta_x| \leq 7\ \mu\mathrm{rad}$ $|\Delta\theta_y| \leq 9\ \mu\mathrm{rad}$ $|\Delta L/L| < 0.11\ \%$

Check with the Restgas Target

Scenarios with beam divergence only

- Determination of beam divergence $(p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c)$





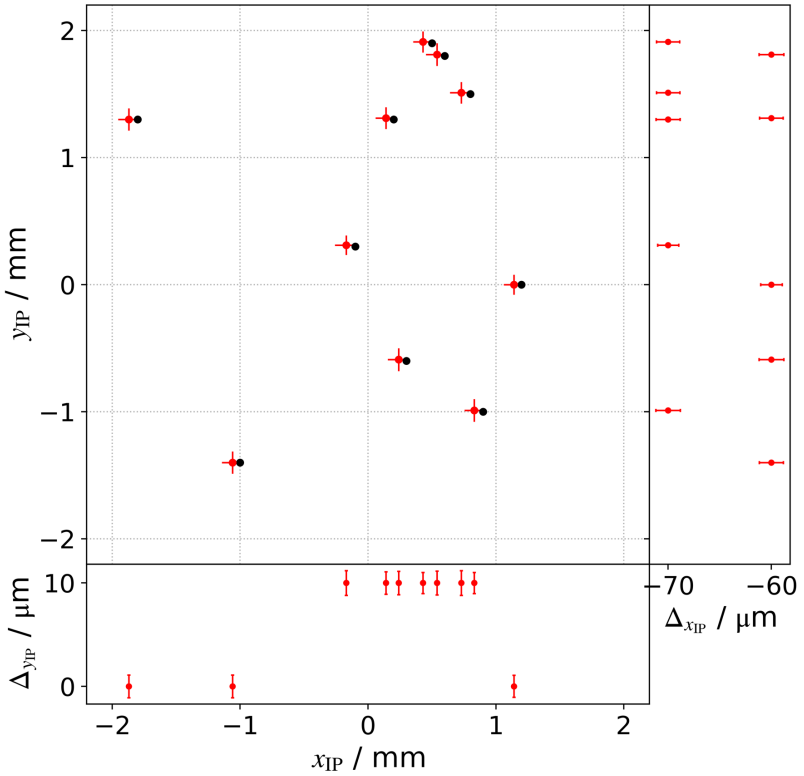
⇨ Maximum deviation after taking into account :

$|\Delta \mathrm{div}_x| \leq 40\ \mu\mathrm{rad}$ $|\Delta \mathrm{div}_y| \leq 45\ \mu\mathrm{rad}$ $|\Delta L/L| < 0.70\ \%$

Comparison w/o Restgas (Realistic Scenarios $p = 4.06\ GeV/c$)

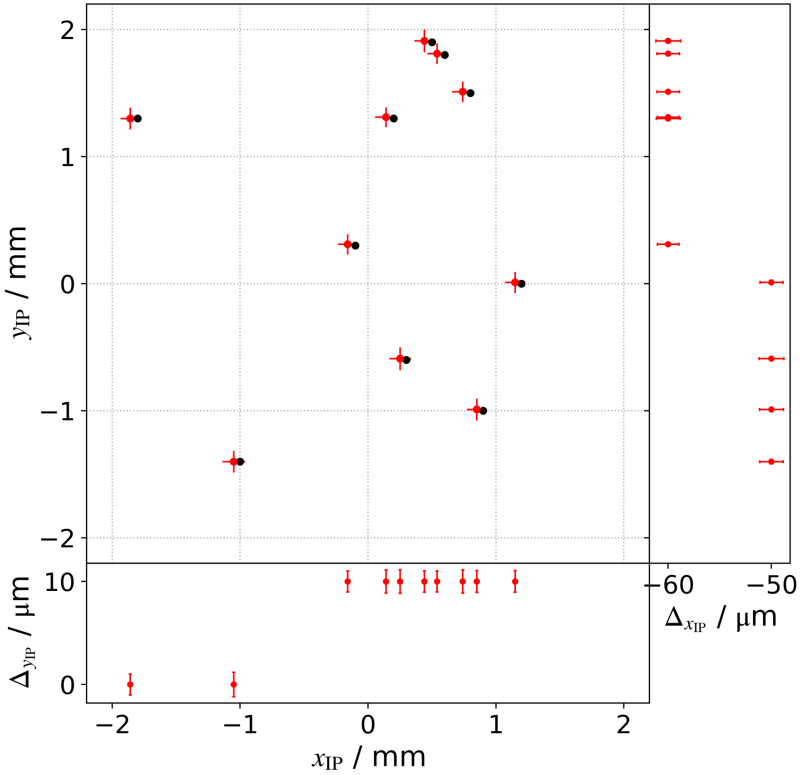
IP shift

With Restgas



⇨ Maximum deviation:
$|\Delta x| \leq 70\ \mu m$ $|\Delta y| \leq 10\ \mu m$

Without Restgas

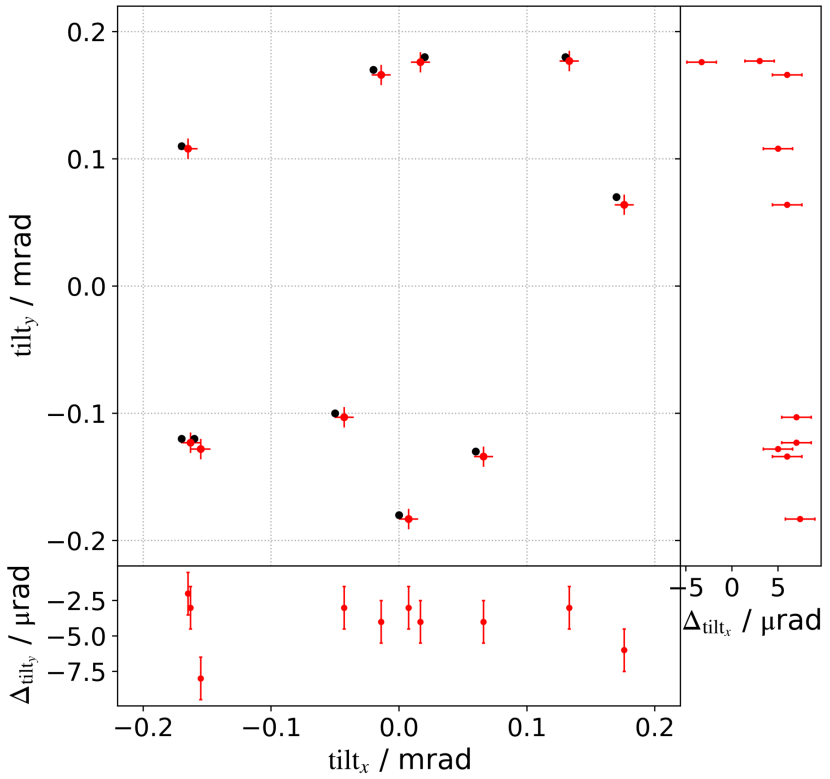


⇨ Maximum deviation:
$|\Delta x| \leq 60\ \mu m$ $|\Delta y| \leq 10\ \mu m$

Comparison w/o Restgas (Realistic Scenarios $p = 4.06\ GeV/c$)

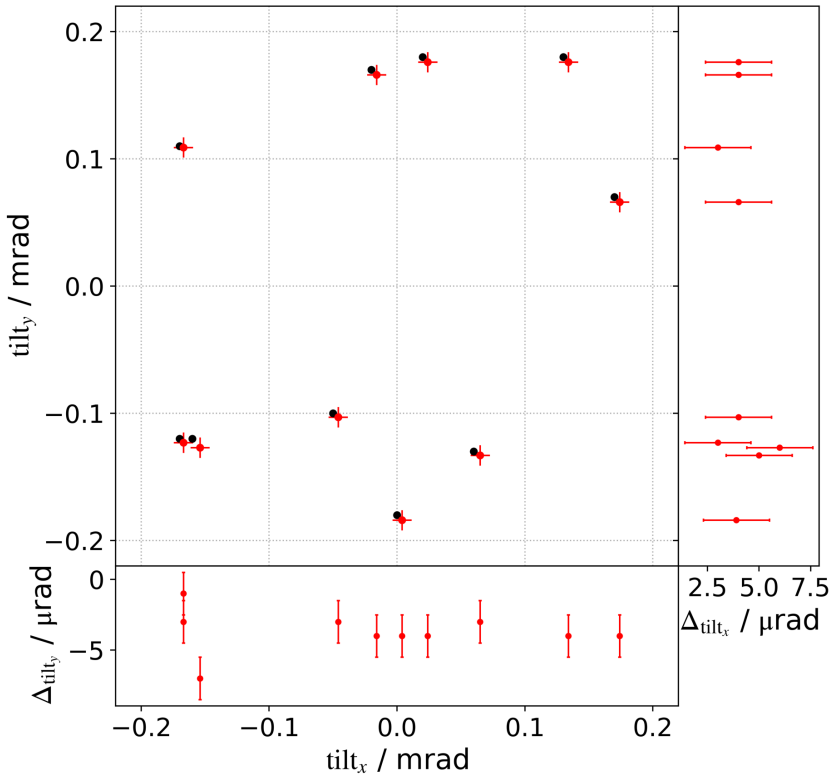
Beam tilt

With Restgas



⇨ Maximum deviation:
$|\Delta\theta_x| \leq 6\ \mu rad \quad |\Delta\theta_y| \leq 7\ \mu rad$

Without Restgas

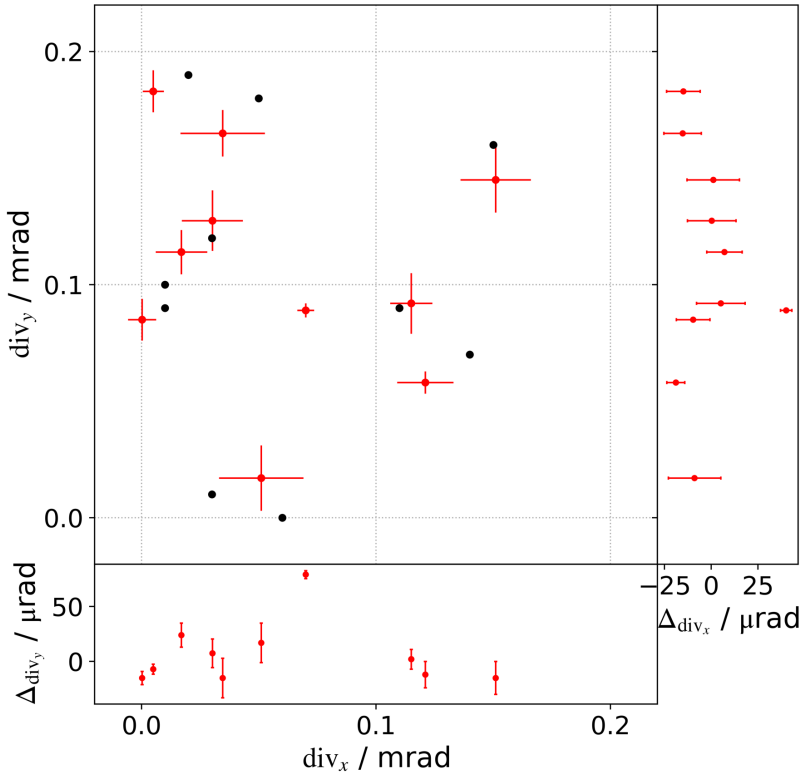


⇨ Maximum deviation:
$|\Delta\theta_x| \leq 8\ \mu rad \quad |\Delta\theta_y| \leq 8\ \mu rad$

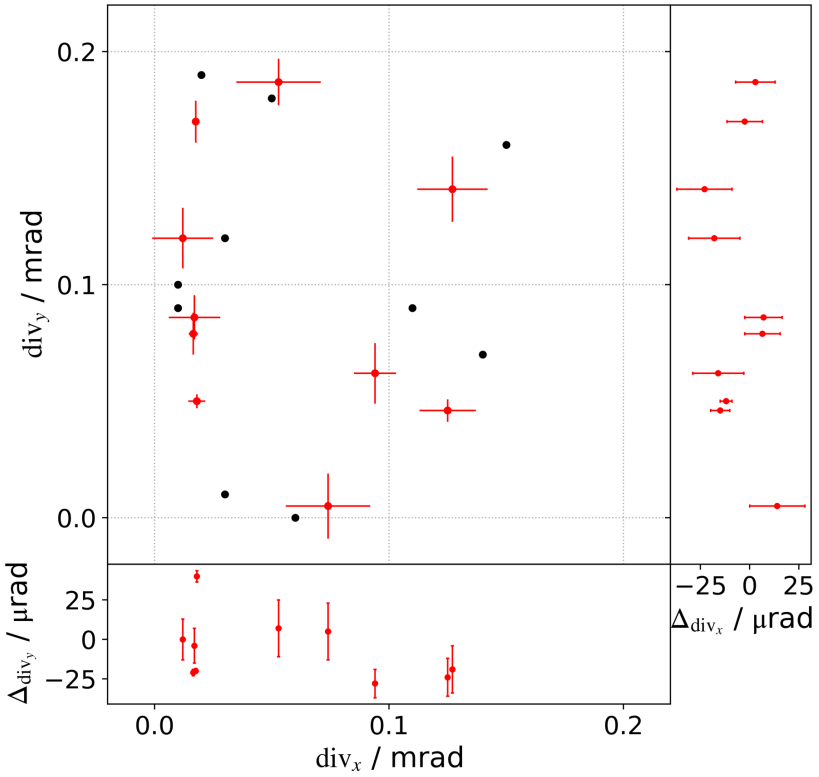
Comparison w/o Restgas (Realistic Scenarios $p = 4.06\ GeV/c$)

Beam divergence

With Restgas



Without Restgas



⇨ Maximum deviation:
$|\Delta \mathrm{div}_x| \leq 40\ \mu\mathrm{rad}$ $|\Delta \mathrm{div}_y| \leq 80\ \mu\mathrm{rad}$

⇨ Maximum deviation:
$|\Delta \mathrm{div}_x| \leq 25\ \mu\mathrm{rad}$ $|\Delta \mathrm{div}_y| \leq 40\ \mu\mathrm{rad}$